

Machine learning is a subset of artificial intelligence focused on building applications that learn from data.

1. Supervised Learning

In supervised learning, algorithms are trained using labeled datasets where the input data is paired with target outputs.

2. Cost Functions and Loss

The Mean Squared Error (MSE) cost function is defined as:

$$\text{MSE} = (1/n) * \sum((y_{\text{pred}_i} - y_{\text{true}_i})^2)$$

3. Gradient Descent Optimization

Gradient descent minimizes the cost function by iteratively adjusting model weights w in the negative direction of the gradient.

$$w_{\text{new}} = w_{\text{old}} - \alpha * \text{grad}(J(w))$$

where α represents the learning rate parameter. A learning rate that is too high causes divergence, while a rate that is too low causes slow convergence.

Page 2: Deep Learning & Neural Network Architectures

1. Perceptron and Multi-Layer Perceptrons (MLP)

An artificial neuron computes a weighted sum of inputs plus a bias term, followed by an activation function.

$$z = \sum(w_i * x_i) + b$$

$$a = \sigma(z)$$

Common activation functions include:

- ReLU (Rectified Linear Unit): $f(x) = \max(0, x)$, which mitigates vanishing gradients.
- Sigmoid: $f(x) = 1 / (1 + \exp(-x))$, useful for binary classification probabilities.
- Softmax: converts raw logits into a normalized probability distribution across multiple classes.

2. Backpropagation

Backpropagation computes the gradient of the loss with respect to each weight using the mathematical chain rule.

Page 3: Model Evaluation and Regularization Techniques

1. Bias-Variance Tradeoff

- High Bias leads to underfitting: the model is too simple and fails to capture underlying data patterns.
- High Variance leads to overfitting: the model memorizes training noise and generalizes poorly to unseen data.

2. Key Classification Metrics

- Precision: $TP / (TP + FP)$ · proportion of predicted positives that are truly positive.
- Recall (Sensitivity): $TP / (TP + FN)$ · proportion of actual positives that were correctly identified.
- F1-Score: harmonic mean of Precision and Recall: $2 * (Precision * Recall) / (Precision + Recall)$.

3. Regularization Strategies

- L2 Regularization (Ridge / Weight Decay): adds sum of squared weights $\lambda * \sum(w_i^2)$ to the loss function to penalize large weights.
- Dropout: randomly deactivates a fraction p of hidden units during training to prevent co-adaptation of features.